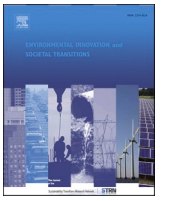
Environmental Innovation and Societal Transitions 61 (2026) 101167



Contents lists available at ScienceDirect

Environmental Innovation and Societal Transitions

journal homepage: www.elsevier.com/locate/eist



Network organizational capabilities for scaling co-created sustainability solutions: The Nordic countries as a testbed

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ARTICLE INFO

Keywords:
Co-creation
Sustainability transitions
Scaling up
Scaling out
Nordic model

ABSTRACT

The Nordic countries are widely recognized for their democratic governance, decentralized authority, and trust-based participatory cultures, which are ostensibly conditions that provide fertile grounds for co-created sustainability transitions. However, while such environments might improve the successful design and implementation of niche-level, locally embedded co-created solutions, they do not necessarily guarantee their subsequent scaling. This article examines why some sustainability initiatives are scaled while others do not, despite being embedded in similarly favorable governance conditions. Drawing on the extant literature, we theorize three interdependent network organizational capabilities that underpin the scaling of co-created sustainability solutions: stakeholder anchorage (the alignment and sustained commitment of key actors and legitimacy networks), programmatic integration (the embedding of co-created initiatives within reflexive scaling policies, strategies, and operational frameworks), and resourcing capacity (the institutional ability to mobilize and sustain financial, material, and in-kind resources). We test this framework through a comparative analysis of four Nordic cases from Denmark, Finland, Norway, and Sweden, which together provide a controlled governance setting for examining how the presence (or absence) of these capabilities shapes scaling outcomes. The findings demonstrate that while each capability addresses distinct barriers to scaling, none is sufficient on its own. Rather, scaling success depends on their mutual interdependence and cumulative layering, which jointly generate the relational and institutional capacity necessary for moving co-created sustainability solutions from local experimentation to upward institutional uptake and outward expansion.

1. Introduction

Co-creation has emerged as a promising approach to advancing sustainability transitions by enabling innovative, context-sensitive responses to local challenges. It mobilizes the collective intelligence, resources, and problem-solving capacities of relevant and affected stakeholders, fostering collaborative efforts that address complex sustainability issues from the ground up (Ansell et al., 2022; Kodama, 2015; Stevens and Agger, 2017). When extended beyond their initial local settings, co-created solutions can contribute to sustainability transitions by being diffused to new organizational or geographic contexts and by becoming embedded in broader governance and

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https://doi.org/10.1016/j.eist.2026.101167
Received 14 November 2025; Received in revised form 6 May 2026; Accepted 24 June 2026
Available online 27 June 2026
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policy frameworks, for example through integration into higher-level strategies, regulations, or public programs led by national governments, international organizations, or other authoritative actors. In this article, scaling is conceptualized as a multi-dimensional process through which locally co-created solutions move beyond isolated, localized settings through the combined and interacting dynamics of horizontal diffusion (scaling out or replication) and vertical institutional embedding (scaling up or upward policy integration). While analytically distinguishable, these dimensions may co-evolve as part of broader sustainability transition trajectories (Lam et al., 2020; Moore et al., 2015). Scaling up can enhance the legitimacy, visibility, and standardization of co-created solutions by embedding them in authoritative policy frameworks, thereby improving the conditions for their subsequent uptake and replication across other localities. Similarly, scaling out through diffusion and replication across multiple contexts may reinforce scaling up by demonstrating broader applicability, in effect garnering political support and creating pressure for institutionalization within higher-level governance and policy systems (Palmié et al., 2023; Vreugdenhil et al., 2010; Wigboldus et al., 2016).

According to the extant literature, the success of co-creation is in large part conditioned by the broader political-administrative context (Torfing et al., 2024). Public governance paradigms differ in their institutional openness and capacity to support sustained interactions between state and non-state actors (Osborne, 2010), some of which are ostensibly more conducive to successful collaborative innovation by creating inclusive spaces for active local actor involvement. However, evidence from practice suggests that even in public governance paradigms highly conducive to successful collaboration, niche-level or localized co-created sustainability initiatives are not guaranteed to scale and, in turn, gain transformative impact (Ansell and Torfing, 2015; Conrad, 2015; Turnheim et al., 2015; Zia et al., 2015). In part, this is because co-creation is frequently observed to be designed in response to specific local problems, needs, and conditions, which makes its negotiated outcomes deeply embedded in the specific social, institutional, and material configurations from which they emerge (Chen-Florea et al., 2026). Thus, the very strengths that make co-created initiatives effective locally can become liabilities when attempts are made to scale them. This raises the central research question of the article: *How do network-level collaborative capacities and dynamics influence whether and how co-created sustainability solutions scale beyond their localized context?*

We address this gap by proposing that successful scaling depends on the development of three mutually interdependent network organizational capabilities: stakeholder anchorage, programmatic integration, and resourcing capacity. Stakeholder anchorage refers to the effective alignment and sustained engagement of key actors in dynamic interaction, ensuring both internal legitimacy and external support in emergent collaborative processes (Emerson et al., 2012; Shulla et al., 2020; Sørensen and Torfing, 2003). Programmatic integration captures the embedding of co-created initiatives into reflexive plans, policies, or institutional templates that formalize scaling pathways through adaptive governance (Ansell et al., 2022; Chliova and Ringov, 2017; Ciulli et al., 2022). Resourcing capacity encompasses the organizational capacity to access financial and in-kind resources necessary to maintain and expand collaborative initiatives (Koppenjan and Enserink, 2009; Menon et al., 2024). These three capabilities, we find, are not independent variables but mutually reinforcing capabilities that interact to generate a collaborative network's overall scaling capacity, enabling co-created sustainability initiatives to move beyond isolated local experiments to become durable, scalable transitions managed or governed across scales.

The Nordic governance context provides a methodologically robust setting for examining the interactive complementarities among the three organizational capabilities. The Nordic countries are widely recognized as exemplary environments for collaborative governance, marked by strong democratic institutions, political decentralization, high levels of public trust, low corruption, and an active civil society and business sector (Greve et al., 2016; Krogh et al., 2022). These shared institutional features minimize cross-country variation in the structural preconditions typically associated with successful co-creation, thereby offering a controlled empirical context for systematically comparing how organizational capabilities shape scaling outcomes. The analysis draws on four Nordic case studies of collaborative networks that all qualify as successful co-creation initiatives, meaning that they accomplished their collaboratively defined objectives within the intended project scope, achieved implementation of the agreed outputs, and either concluded as planned (as in the case of time-bound pilots or policy experiments) or became consolidated and stabilized within their locally-bounded contexts. However, only one has managed to scale beyond their initial localized context. This variation enables an analytical focus on the conditions that differentiate co-created initiatives that expand from those that remain localized.

The remainder of the article is structured as follows. Section 2 reviews existing literature on co-creation and the scaling of sustainability innovations, identifying key conceptual gaps. Section 3 develops the theoretical framework, introducing the three mutually interdependent conditions necessary for scaling co-created sustainability transitions. Section 4 outlines the methodology and data collection strategy used to assess these conditions across four Nordic cases. Section 5 presents the case analyses, examining how the presence, absence, and interaction of the three conditions shape scaling outcomes. Section 6 concludes by synthesizing the main findings, discussing implications for theory and practice, and outlining directions for future research.

2. The making and scaling of co-created sustainability transitions

The escalating climate crisis and its pervasive, cross-sectoral impacts have made it increasingly imperative to mobilize both state and non-state actors in the pursuit of effective sustainability solutions. In response to these urgent challenges, a variety of governance strategies have emerged. Traditional approaches have relied primarily on market-based mechanisms, which stimulate innovation through competition and profit incentives, or hierarchical policy instruments, which impose regulatory limits, bans, and mandates to correct unsustainable practices (Jessop, 2016). While both governance strategies have generated important results, their limitations in addressing multifaceted and interdependent sustainability challenges have become increasingly apparent. Against this backdrop, the literature on collaborative governance and collective action, including research on co-creation, has advanced a polycentric understanding of governance that emphasizes cooperation and coordination across multiple centers of authority (Ba et al., 2024; Ostrom,

2010). Collaborative governance mobilizes diverse public, private, and civic actors to take collective and complementary action across levels and sectors. This horizontal, network-based mode of governance prioritizes joint problem-solving and mutual learning, which thereby expands the available tools in the governance repertoire for tackling "wicked" sustainability problems (Alford and Head, 2017). The constellation of participating stakeholders involved in these collaborative processes may be described as collaborative networks, referring to inter-organizational governance arrangements in which coordination is achieved primarily through negotiated interaction and relational work, rather than through hierarchical authority or market mechanisms (Chen and Hustad, 2025).

Within this broader collaborative turn, co-creation has emerged as a distinct approach to collaborative governance aimed at addressing complex sustainability challenges. Co-creation is defined as a governance process in which diverse yet interdependent public, private, and civic actors are actively and continuously involved across multiple—and sometimes all—functional segments of the innovation lifecycle, including problem framing, solution design, experimentation, implementation, and evaluation (Ansell et al., 2022; Hofstad and Vedeld, 2021). Through such continuous engagement, co-creation processes foster joint ownership and shared accountability across sectoral boundaries, enabling the development of innovative solutions that are not only technically sound but also socially legitimate and politically viable (Ansell et al., 2022). This inclusive engagement across the innovation lifecycle supports consensus-building, mobilizes stakeholder interests, and facilitates negotiated compromises that, in turn, reduce latent resistance and institutional friction that might otherwise obstruct the implementation or long-term sustainability of co-created solutions (Ciulli et al., 2022).

While co-creation is ideally grounded in ideals of inclusiveness and equal partnership (Gray et al., 2022), in practice collaborative networks are invariably characterized by power asymmetries. Participants differ in their formal mandates, resources, expertise, and positional authority, which in turn shapes the scope of their roles and the extent of their influence within the co-creation process (Hafer et al., 2022; Purdy, 2012; Ran and Qi, 2018). Levels of engagement and investment may therefore vary across actors, reflecting both their capacities and their strategic interests in the initiative. Such uneven participation is not necessarily problematic, provided that actors retain meaningful opportunities to contribute, that decision-making processes remain transparent and inclusive, and that asymmetries are acknowledged and actively managed rather than obscured (Ostrom, 2009). Under these conditions, differentiated roles and influence can enhance rather than undermine co-creation by allowing actors to contribute in ways aligned with their specific competences and responsibilities. At the same time, power asymmetries can sometimes generate fault lines within collaborative networks if severe, which consequently block meaningful participation and ultimately constrain creative and innovative problem-solving if left unaddressed (Ran and Qi, 2018).

Consequently, co-created solutions to collective sustainability challenges can yield promising niche-based, socio-technical innovations, introducing novel configurations of ideas, actors, and practices within specific governance and community settings. Importantly, their innovative quality is not necessarily tied to technological novelty per se, but to contextual novelty. It may involve the adaptation and recombination of existing solutions, technologies, and practices to make them effective within new local or institutional contexts, for example by transferring proven models from other sectors or regions, reconfiguring known technologies to address local needs, or reframing sustainability problems in ways that unlock new forms of cross-sectoral collaboration (Mulgan, 2019; Voorberg et al., 2015).

2.1. *The sources of failure and success in co-creation*

While collaborative governance and co-creation are often promoted as adaptive and inclusive alternatives to hierarchical or market governance, their self-regulating and consensus-oriented nature also makes them vulnerable to distinct challenges (Chen and Hustad, 2025). Because collaborative networks rely on voluntary engagement and the coordination of diverse actors with differing interests, values, and institutional affiliations, they are inherently fragile and contingent (Klijn et al., 2025). Misaligned goals or conflicting priorities can generate friction and erode collective commitment[1] (Ansell and Gash, 2008). Such tensions are often amplified by interpersonal distrust, reputational concerns, or reluctance to share knowledge, which inhibit open communication and mutual learning (Emerson and Nabatchi, 2015). Above all, in contexts marked by weak relational histories, collaboration risks remaining transactional rather than genuinely co-creative, undermining the capacity for innovative problem-solving.

Collaborative processes are also prone to power and resource asymmetries, whereby dominant actors may shape agendas or marginalize weaker participants, jeopardizing inclusiveness and legitimacy (Chen and Hustad, 2025; Newig et al., 2018). These asymmetries often mirror external institutional barriers such as rigid legal mandates, siloed bureaucracies, or unclear administrative procedures. As a result, co-creation arenas may evolve into parallel or semi-autonomous structures, raising questions of accountability and blurred lines of authority between collaborative networks and formal political-administrative bodies (Steen et al., 2018). Finally, without adequate financial, organizational, or cognitive resources, many co-creation initiatives struggle to sustain engagement over time, leading to participant fatigue, diminished capacity, and unmet expectations (Bryson et al., 2006).

Nevertheless, the literature identifies several enabling conditions that increase the likelihood of successful co-creation. These include high levels of interpersonal and institutional trust, inclusive participatory channels, and strong relational infrastructures that lower transaction costs for collective decision-making (Ansell and Gash, 2018; Emerson and Nabatchi, 2015). A critical factor in this

[1] While collaborative networks are often voluntary in nature, they can also be partially mandated by public authorities or policy frameworks (Krogh, 2022). However, even in such cases, elements of voluntarism and discretion remain part and parcel of collaborative governance. If every aspect of collaboration were formally mandated, it would eliminate the very agency, deliberation, and creative problem-solving capacity that define collaboration in the first place, thereby voiding the logic of collaborative governance altogether.

regard is the presence of a facilitative network leader or intermediary who convenes, translates, and brokers relationships across sectoral boundaries (Hofstad et al., 2022, 2023). Such network leaders play a vital role in articulating a shared vision, managing knowledge and resource flows, maintaining trust among stakeholders, and linking the initiative to external networks, funders, and policy platforms. In addition, governmental actors must also play a pivotal role as orchestrators of collaborative experimentation within multi-level governance systems (Bulkeley and Broto, 2013; Hofstad et al., 2023), whereby they absorb some of the coordination costs that would otherwise deter private and civic participation (Ansell et al., 2022).

These enabling conditions are especially salient in the Nordic governance context, widely regarded as one of the most fertile institutional environments for collaborative governance and co-creation (Øjvind et al., 2024; Røiseland, 2022; Røiseland et al., 2024). The Nordic countries combine effective multi-level governance with strong democratic traditions, high public trust, and state regimes open to public–private–civic collaboration (Greve et al., 2016; Torfing et al., 2019, 2024). These enabling governance contexts provide fertile grounds for public actors' willingness and ability to participate in and convene local co-creation processes in pursuit of collective sustainability goals (Ansell et al., 2022). These same structural conditions also reinforce relational enablers of collaboration by institutionalizing participation, fostering trust, and creating the administrative space and normative expectations necessary for cross-sectoral engagement (Greve et al., 2016). In effect, the Nordic model nurtures the underlying relational and structural capacities upon which collaborative governance depends.

2.2. *Latent tensions between co-creation and scaling*

Many co-created sustainability initiatives are deliberately designed to address place-based problems within bounded contexts, where local consolidation and stabilization of collaborative practices may be both analytically meaningful and normatively desirable. Thus, scaling is not an inherent requirement for the success of co-creation.[2] However, when the ambition is to extend co-created solutions beyond their original local setting, such that they may contribute to broader sustainability transitions, the challenges encountered change fundamentally. In such cases, co-creation initiatives must move beyond localized experimentation and develop the capacity to transfer, adapt, or embed their knowledge, practices, and institutional learning in new contexts. Importantly, successful scaling does not imply that the entire co-creation process is repeated from scratch; rather, it presumes that earlier investments in collaboration can be leveraged to enable expansion without incurring the full costs of initial experimentation. While the various contextual and organizational factors outlined in the prior section may foster effective co-creation within the boundaries of a given project, these do not inherently guarantee that such initiatives will be scaled beyond their original context.

Instead, the broader innovation literature issues additional requirements, reflected in several established models for scaling. Phased growth frameworks emphasize a stepwise progression from pilot testing to early adoption, and thereafter full-scale implementation (Latypov et al., 2024; Seelos and Mair, 2017). Market-led diffusion strategies focus on profitability and competitive advantage as drivers of uptake, while state-led scaling models rely on top-down policy diffusion, regulatory mandates, or public investment to institutionalize new practices (Mulgan, 2019; Rogers, 1995). Across these approaches, the shared, underlying assumptions is that the innovation in question is relatively standardized, that it will be deployed within a stable and predictable institutional environment, and that there are clear economic or policy incentives for adoption (Westley and Mcconnell, 2010). Scaling is thus understood largely as a function of de-risking the innovation, securing investment, and leveraging supportive infrastructure. While these models have often proven effective in certain technological, commercial, and policy domains, they offer limited guidance for innovative, co-created sustainability solutions where success depends on the negotiated alignment of diverse stakeholders and adaptation to highly context-specific governance environments.

In local co-created settings, the very features that make collaborative problem-solving effective—its embeddedness in local governance arrangements, responsiveness to specific socio-ecological contexts, and tailoring to a particular constellation of stakeholders—can become liabilities when scaling (S. B. Page et al., 2023). Such niche-level, localized solutions often reflect parochial interests, negotiated compromises, and context-specific trust relationships that may not be translated elsewhere. These difficulties mirror a core insight from the policy experimentation and piloting literature. Pilots are frequently used as time-limited, bounded tests of novel solutions that enable learning under uncertainty and allow actors to explore innovation–context interactions in protected "safe spaces" before committing to wider roll-out (Mei and Liu, 2014; Nair, 2021). Yet this literature also emphasizes that pilot success is often contingent on exceptional and temporary conditions—concentrated resources, unusual levels of attention, and supportive coalitions—that reduce risk and uncertainty but also limit representativeness and, in turn, transferability (Nair and Howlett, 2016; van Buuren and Loorbach, 2009). Accordingly, the translation of experimental success into broader adoption is frequently obstructed by well-documented diffusion and institutionalization hurdles, such as weak learning mechanisms, lack of institutional embedding, poor timing, and "wait-and-see" attitudes among prospective adopters facing uncertain responsibility and financial risk (Vreugdenhil et al., 2010).

In effect, experimental initiatives may remain "innovations in isolation" unless deliberate efforts are made to connect them to

[2] In some instances, overextending the ambition towards scaling might even be detrimental to co-creation initiatives, especially if participating stakeholders lack the resources, motivation, or capacity for follow-through (Nair, 2021). From this perspective, the consolidation and stabilization of co-created solutions within their original context—sometimes described as "scaling deep" may be the preferred option (André & Pache, 2016). For the purposes of this article, we disambiguate scaling deep from our conceptualization of scaling up or out due to their different analytical foci. The emphasis placed on scaling up or out is connected to the broader claim that scaling is necessary for co-created solutions to become catalyst for broader sustainability transitions, as they cannot be expected to engender systematic change through small isolated changes.

broader legitimacy networks and institutional routines beyond the pilot setting. Thus, without deliberate adaptation for scaling, such co-created sustainability solutions may fail to resonate beyond their original setting. Additionally, then, there seems to be a need for network leadership that can mobilize resources, broker relationships in the local context, and institutionalize co-created outcomes through upward or outward bargaining to change the institutional context conditioning the co-created sustainability solutions (Conrad, 2015; Torfing et al., 2024). The question, then, is: what specific concrete tasks and roles must local management and network leadership perform to enable the scaling of co-created sustainability initiatives?

3. The mechanisms of scaling co-created sustainability transitions

The question of why some innovations scale while others remain confined to isolated and localized has long preoccupied the innovation literature. A bridging perspective can be found in the dynamic capabilities perspective, which emphasizes that organizations succeed in scaling not merely because they possess a static stock of resources, but because they can build, reconfigure, and renew organizational competences in response to changing environments (Teece, 2009; Teece and Pisano, 1994). Dynamic capabilities represents the nexus between cognitive processes, organizational routines, and intentional agency, which manifest as organizational capacities to intentionally sense opportunities for scaling or expansion, seize them through the mobilization of strategic commitments among relevant organizational stakeholders, and, in turn, reconfiguring organizational resources and relations to adapt to these new market contexts. In sum, (dynamic) capabilities are understood as higher-order organizational capacities that enable purposeful adaptation in changing environments, hereunder (market) scaling and expansion (Teece, 2009)

We argue that an analogous perspective is needed to understand why some co-created sustainability solutions achieve scaling while others do not. Collaborative networks, like firms, face the organizational challenge of mobilizing and reconfiguring relations, strategies, and resources under conditions of uncertainty and contestation when attempting to scale. However, unlike firms, their effectiveness depends not on hierarchical command but on relational work, legitimacy, and institutional embedding across diverse stakeholders (Chen and Hustad, 2025; Emerson et al., 2012). Importantly, adopting a dynamic capabilities perspective does not imply a unidirectional or mechanical causal relationship between (network) organizational capabilities and scaling outcomes. Rather, capabilities and scaling ambitions are understood as recursively related. While organizational capabilities condition whether scaling can be realized, they rarely emerge spontaneously; they are typically cultivated when actors articulate an explicit scaling ambition.[3]

From this vantage point, we define scaling capacity as *the composite and evolving capacity of a collaborative network to extend* and adapt *a co-created solution beyond its initial, localized context through horizontal diffusion (scaling out) and vertical institutional embedding (scaling up), while preserving its core value proposition* (Ciulli et al., 2022; Hofstad et al., 2026; Lam et al., 2020; Wigboldus et al., 2016). Building on insights from the sustainability governance, innovation management, and collaborative governance literature, we postulate that scaling capacity in co-created settings rests on the combined presence of three network organizational capabilities: *stakeholder anchorage, programmatic integration,* and *resourcing capacity* (see Fig. 1). Scaling capacity is thus relationally constituted as it emerges through the evolving configurations of institutional linkages, political alignments, and resource channels that the collaborative network brokers within and beyond its original context.

Stakeholder anchorage refers to the degree to which a collaborative network and its co-created solutions are embedded in durable relationships that confer legitimacy, credibility, and sustained commitment among relevant actors. Durability in this sense does not imply permanent or indefinite relationships, but persistence through the critical phases required for scaling and the subsequent process of consolidation. In the context of sustainability transitions, stakeholder anchorage requires more than mobilizing stakeholders during the design and implementation phases (Emerson and Nabatchi, 2015). It entails the ongoing alignment of interests and shared ownership of objectives with an explicit orientation toward scaling (Ansell and Torfing, 2015). Beyond cultivating strong ties among project partners, there is a need for deliberate efforts to bridge outward beyond the initiative's initial local network boundary to external or higher-level stakeholders whose support is necessary for scaling (Conrad, 2015). These external stakeholders may include policymakers operating at higher administrative levels, public agencies in other jurisdictions, potential funders or investors, organized community groups in prospective adoption sites, and future end-users (S. Page, 2010).

Without robust stakeholder anchorage that extends beyond the local context, the legitimacy and buy-in achieved during co-creation may fail to travel to other settings. Without robust stakeholder anchorage, the legitimacy and buy-in achieved in the local context may not travel to other settings, leaving the innovation vulnerable to political resistance, declining participation, or erosion of credibility when introduced elsewhere (Vedeld et al., 2021). Anchorage is therefore evidenced by the credible commitment of external stakeholders to support scaling efforts which may take the form of binding endorsements or declarations of intent, alignment with policy agendas or strategic priorities, participation in governance or steering arrangements, or the provision of tangible resources such as funding, expertise, infrastructure, or access to implementation sites. In this sense, stakeholder anchorage functions as a relational bridge that enables co-created solutions to move beyond their local constituency and gain the legitimacy required for expansion across contexts.

Stakeholder anchorage also involves a governance dilemma regarding scope (Ansell and Torfing, 2021b; Jessop, 2020). A narrow anchorage, built on a small set of strongly committed stakeholders, can provide stability and potentially support the local consolidation of the co-creation solution (André and Pache, 2016). However, it simultaneously risks lock-in effects, as early negotiated interests may

[3] Conversely, in the absence of such ambitions, collaborative networks tend to consolidate around—or be constrained by—localized problem-solving and the organizational capacities required for scaling, in turn, remain underdeveloped. Scaling capacity should therefore be understood as an emergent property of iterative interactions between strategic intent and organizational capability-building over time.

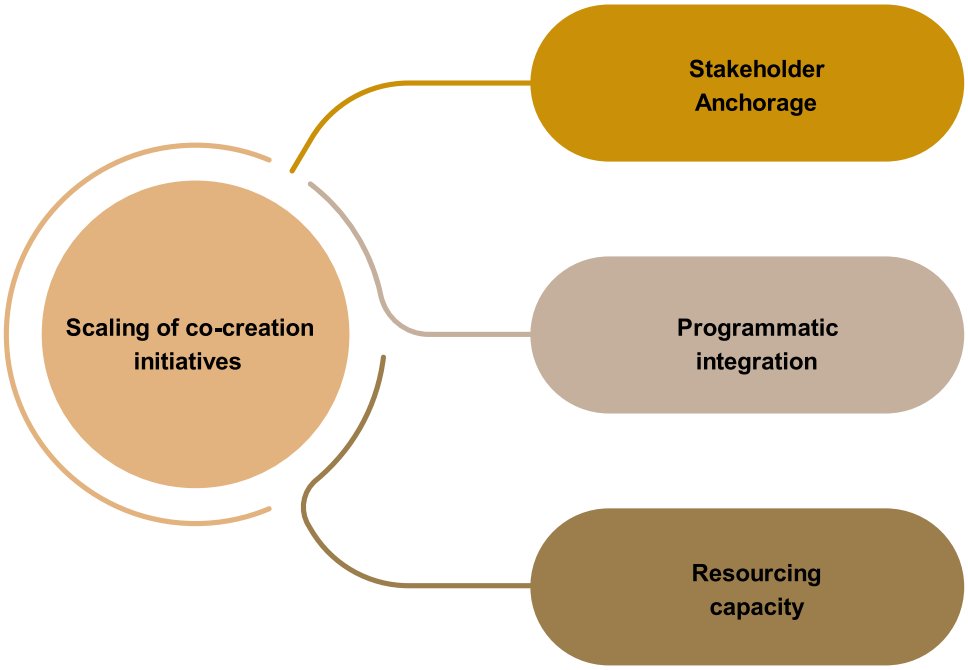


Fig. 1. The three organizational capabilities creating scaling capacity.

constrain broader adaptation and subsequent scaling. A broad anchorage, by contrast, enhances legitimacy and resource diversity but increases coordination complexity and conflicting priorities that might jeopardize the take-off of the localized co-creation initiative altogether. The appropriate scope of anchorage is thus context-sensitive, shaped by scaling ambitions and governance environments (Torfing et al., 2024). Narrow anchorage may suffice for niche scaling, while broader anchorage is indispensable for cross-sectoral or multi-jurisdictional uptake.

Programmatic integration denotes the extent to scaling ambitions are embedded within the policies, plans, and operational routines of the collaborative network and its adaptive governance systems. It transforms otherwise location-bound or experimental projects into enduring, institutionalized practices, providing both a strategic direction and a concrete pathway for expansion (Ansell and Torfing, 2021b). Following Kingdon (2014), this may involve codifying innovations into strategic plans, framing them as viable solutions to already recognized policy problems, and aligning them with prevailing political priorities and institutional agendas at the governance levels targeted for scaling. It may further entail embedding co-created solutions within formal planning, budgeting, performance, or accountability systems of the organizations or authorities expected to adopt or mainstream them, thereby positioning the innovation to be taken up when policy windows open and decision venues become receptive.

In parallel, programmatic integration helps develop the organizational mechanisms that enable collaborative networks to navigate shifting incentives, interests, and capacities as sustainability solutions expand beyond their initial context. This adaptive process must facilitate the recalibration of benefits, costs, and commitments among stakeholders, allowing for the maintenance of dynamic yet stable compromises (Kooiman, 2003; Patterson, 2021). For example, when a locally co-created solution is introduced into a new jurisdiction or policy domain, public authorities may face additional regulatory or budgetary constraints, private actors may encounter altered risk–return profiles, and civil society participants may experience changes in their influence or workload. Programmatic integration implies creating a framework within which such shifts can be renegotiated—by adjusting roles, redistributing responsibilities, modifying implementation timelines, or revising cost-sharing and incentive structures—so that actors remain willing and able to participate despite changing institutional or geographical contexts (Ansell et al., 2022; Nair, 2021)(Ansell et al., 2022; Ansell and Torfing, 2021a; Nair, 2021). Programmatic integration thus encompasses both the institutional embedding of co-created solutions as well as their purposeful translation and contextual adaptation as part of broader scaling trajectories.

However, despite the importance of programmatic integration, many co-creation initiatives fail to scale because such considerations are addressed only ex post facto, after the pilot phase has ended, by which point the necessary relational and institutional capacities are difficult to retrofit. Early integration provides stability and direction but risks premature institutionalization that locks in suboptimal designs. Conversely, delayed or gradual integration, while allowing for experimentation and adaptive learning (Ansell et al., 2024), may result in the loss of momentum, stakeholder engagement, or political support needed for institutional uptake (Vedeld et al., 2021). A central challenge, therefore, lies in ensuring that the scaling trajectory of co-created sustainability solutions can navigate and eventually transcend routinized and silo-based administrative structures (Bulkeley and Broto, 2013; van der Heijden et al., 2019). This is because, while localized, niche-level pilot projects may thrive within existing political-administrative boundaries, successful scaling requires deliberate negotiation of these institutional constraints and the crafting of viable pathways for integration within the governance systems and wider institutional environment they are embedded.

Resourcing capacity constitutes the ability of a collaborative network to mobilize, attract, and sustain the financial, material, and in-kind resources required to extend an innovation beyond its original project scope. The concept builds on and extends the cognate notion of mobilization capacity (Menon et al., 2024), defined as the ability of actors to access and extract resources from their networks, but generalizes this capacity from episodic mobilization to long-term, institutionalized capacity at the network level. Thus, whereas mobilization capacity emphasizes immediate access to resources within existing networks, resourcing capacity refers to the collaborative network's ability to institutionalize resource flows, sustain credibility with funders, and, consequently, continually renew its access to financial and in-kind support.

Resourcing capacity is already central to successful co-creation when initial pilots and experimental solutions want to consolidate, ideally through blended and diversified financing streams to ensure that the delivery of collaborative outputs does not terminate if a single, dominant funder withdraws (Ansell et al., 2022; Havemann et al., 2020). However, when co-created solutions seek to scale beyond their original local context and engage a broader set of external stakeholders, the nature of resourcing capacity changes qualitatively rather than merely quantitatively. Scaling introduces new categories of actors—such as additional municipalities, sectoral agencies, private investors, or end-users—who were not necessarily a central part of the initial, localized co-creation processes and whose participation cannot, therefore, be secured solely through prior relational commitments. As a result, resourcing capacity must extend beyond securing short-term project funding to establishing credible, repeatable, and institutionally legible resource pathways that can sustain expansion across organizational and territorial boundaries (Øjvind et al., 2024).

To achieve this, it is necessary to build a credible proof of concept that resonates with funders (Bloom and Chatterji, 2009), cultivating relationships with potential investors or resource partners, and leveraging complementary in-kind contributions, such as specialized expertise, infrastructure, or volunteer labor. Resourcing capacity is therefore critical for ensuring that scaling efforts are not constrained by the depletion of initial funding or human capital, and for enabling the iterative adaptation necessary during expansion (Austin et al., 2006). In practice, many co-creation initiatives are financed as time-bound pilot projects with no resources earmarked for scaling (Røiseland et al., 2024). Without deliberate investment in developing resourcing capacity through relational ties with potential funders and resource partners, subsequent scaling attempts are likely to falter regardless of the initiative's legitimacy or programmatic integration.

Table 1
List of case vignettes.

Name	Country	Description	Scale	Years active	Scope
SkyClean	Vrå, Denmark	The project sought to develop and commercially scale the use of pyrolysis as a technology for carbon capturing and storage, which successfully culminated in the development of a proof of concept—in the form of a viable sustainable business model—that can be sold as modular solutions to farmers, energy companies, and other prospective end-users.	Regional, across multiple municipalities in Northern Jutland	2018-ongoing	50 core participants consisting primarily of Stiesdal (Owner) and various engineers, researchers, and managers involved in testing and scaling the prototype.
Circular Green Blocks	Helsinki, Finland	The project sought to create a circular economy platform that would help companies identify and develop sustainable business models related to sustainable mobility, circular economy, and collaborative consumption, and subsequently scale them upwards or outwards. The project supported many pilot projects, but none discernibly scaled upwards or outwards as the project was terminated once funding was exhausted.	City-municipal	2021–2024	40 core participants across, across 10 pilots. HSY, the Helsinki regional authorities acted as network leaders, collaborating with private circular service providers, researchers, thinktanks, and end-users to devise and test urban circular economy solutions. Many core participants were involved in multiple pilots. End-users in the form of local residents were involved in larger magnitude, but in a limited capacity.
Arena Breivoll	Oslo, Norway	The project sought to provide bring local private stakeholders, public, authorities and citizens for the joint mobilization of innovative ideas, pilots, and resources for sustainable development of the local area with the ambition of scaling them outwards or upwards, although it failed due to a lack of uptake and broader support in terms of resources and funding.	City-municipal	2012–2023	30 core participants, spanning the public, private, and civil society sectors. Citizens and end-users would be variably included in a larger magnitude, but in a limited capacity in relation to a given pilot.
Rundgång	Karlstad, Sweden	The project sought to co-create circular economy initiatives centered around sustainable consumption, in particular a recycling mall, as part of the Climate Neutral Karlstad 2023 initiative. The project never managed to formulate a concrete vision for scaling upwards or outwards, hence it consolidated locally.	City-municipal	2021-ongoing	50 core participants, including 40 small businesses participating in the recycling mall directly. A local mall participated by offering a space for hosting the recycling mall, joined by various civil society organizations and private interest organizations funding the activities.

In sum, the extant literature provides strong theoretical grounds for expecting that *stakeholder anchorage, programmatic integration*, and *resourcing capacity* are critical drivers for scaling co-created sustainability solutions into broader sustainability transitions. Nonetheless, the literature remains fragmented, with scant systematic attention to their complementarities, trade-offs, or cumulative dynamics. Various traces in the extant literature suggest that these network organizational capabilities influence each other. According to Menon et al. (2024), the capacity for network organizations to mobilize resources (resourcing capacity) may be compounded by, among other things, the continuous alignment of motivation among stakeholders (stakeholder anchorage), which can potentially catalyze a virtuous cycle. Conversely, according to the pilot and policy experiments literature, a lack of acceptance or support from key stakeholders (stakeholder anchorage) might be symptomatic of a lack of programmatic integration of scaling ambitions, manifesting as an incomplete "identification of potential future users or application sites and of the nature of what ought to be diffused" (Vreugdenhil et al., 2010, p. 17).

Specifically, within the context of co-created sustainability transitions, Torfing et al. (2024) cautiously argue that governance factors, such as financing access (resourcing capacity) and the cultivation of upward linkages to governments (stakeholder anchorage), "*may combine* to produce positive outcomes in the absence of other factors" (Torfing et al., 2024, p. 5 emphasis added), but maintain that these are theoretically open-ended questions. Taking cues from this call for further research, we may ask if the scaling capacity of co-creation initiatives is simply a function of the relative presence of these capabilities, or whether they operate as necessary and interdependent conditions. We take up this question next in the analysis and discussion sections, where we explore their reciprocal interdependence in light of our empirical cases.

4. Methodology

The theoretical significance of the three network organizational capabilities is examined through four case studies of co-created sustainability transitions in the Nordic countries (see Table 1). The Nordic context provides a methodologically sound setting for comparative analysis, as the countries share broadly similar political–administrative systems, socio-cultural norms, and governance traditions, which constitute factors consistently associated with favorable conditions for co-creation (Greve et al., 2016; Krogh et al., 2022). This shared backdrop effectively controls for general contextual confounders, allowing us to focus on the organizational capabilities that differentiate scaling outcomes. The four cases of collaborative networks qualify as successful co-creation processes in the sense that they have achieved the implementation of a collaboratively developed solution within their initial project scope. However, they vary in whether, and to what extent, these solutions were subsequently scaled beyond the original constellation of stakeholders engaged in the co-creation process, i.e., scaled out to have wider impact in socio-institutional-geographic terms, or scaled up to influence wider policy or institutional frameworks. This variation enables the application of a set-theoretic comparative logic, specifically Mill's method of difference, operationalized within a configurational comparative framework (Rihoux and Ragin, 2009).

Two interrelated methodological rationales guided the case selection. First, the selection is grounded in the implicit hypothesis that all three organizational capabilities are necessary conditions for developing scaling capacity. Following a typical case selection logic (Seawright and Gerring, 2008), the chosen cases thus represent empirically relevant instances in which the hypothesized relationships can be meaningfully observed and tested within comparable governance settings. Second, the case set was designed to capture systematic variation in the degree to which each capability is present. Each case was assessed along a three-level scale—absent, partially present, or present—allowing for a more nuanced examination of how different constellations of capabilities influence scaling outcomes. This comparative makes it possible to analyze both the individual contributions of each capability and their cumulative or interactive effects. Beyond testing for additive relationships, the design also enables exploration of potential interactive complementarities among the three capabilities, that is, whether they operate as mutually reinforcing and co-constitutive conditions rather than as discrete, independent variables.

The four case studies have been conducted based on a shared research protocol, ensuring comparability across cases by standardizing the scope of inquiry, interview procedures, and analytical criteria (see ANONYMIZED). In total, the cases draw on 48 semi-structured interviews (10–14 interviews for each case study) collected between 2023 and 2025, conducted using a snowball sampling strategy (Farquharson, 2005). Each case began with interviews of network leaders, after which the sample was extended to other key actors they had identified (see Appendix 1 for a list of interviewees). This approach ensured the inclusion of all relevant stakeholders and core actors, including those who may not have been formally central to the collaborative network but played critical roles in shaping the process or outcomes. The interview data were complemented by case reports produced for the GOGREEN project (www.gogreen-project.com), which provided additional documentary evidence, contextual background, and triangulation for validating findings (Askola and Chen, 2024; Granberg, 2024; Torfing and Sørensen, 2024; Vedeld and Mccrory, 2024). Together, these sources yielded a rich, multi-perspectival dataset for assessing the presence, absence, and interaction of the three institutional conditions across cases.

The interviews covered the following thematic domains, based on the GOGREEN research protocol: the origins and evolution of the co-creation process, including its initial problem framing and design; the roles, motivations, and interactions of participating stakeholders; governance arrangements, decision-making procedures, and leadership dynamics within the collaborative network; the mobilization and management of financial, material, and in-kind resources; and experiences with, or barriers to, scaling the co-created solution beyond its initial context (see Ansell et al., 2023 for additional details).

Each case study was analyzed using an abductive analytical strategy (van Hulst and Visser, 2024), structured in three iterative steps. First, we developed a sensitizing, proto-theoretical framework derived from the scaling, innovation management, and collaborative governance literatures. This framework specified three network organizational capabilities—stakeholder anchorage, programmatic integration, and resourcing capacity—which served as an initial deductive filter that oriented data collection, coding, and

cross-case comparison (Thompson, 2022).

Second, we conducted an inductive within-case analysis that was deliberately constrained by this deductive scaffold. In practice, this meant that we coded the empirical material for concrete manifestations of each capability while remaining attentive to how these capabilities were enacted, contested, and maintained in context (see, for example, Alvesson and Kärreman, 2007; Tavory and Timmermans, 2014). In effect, we did not assume a one-to-one mapping between empirical events and analytic categories; rather, we used the capability framework to structure systematic comparison while allowing case material to specify how the capabilities took form in different organizational and institutional settings (see Appendix 2 for the operationalization schema).

Third, we leveraged cross-case comparisons to generate emergent insights into the relational dynamics linking the three capabilities within co-creation processes. This step focused on identifying recurrent patterns of interaction and sequencing. For example, we examined how shifts in stakeholder commitment affected prospects for institutional embedding and resource mobilization, and how changes in resourcing or programmatic integration fed back into stakeholder anchorage. These iterative comparisons informed refinement of both (1) the empirical meaning of each capability in co-creation settings and (2) the theorized relationships among them, yielding a more context-sensitive account of how combinations of capabilities condition scaling trajectories.

5. Analysis

The analysis proceeds stepwise, moving from the cases characterized by fewer and more weakly developed organizational capabilities to cases in which a greater number of capabilities are present and mutually reinforcing (see Table 2). This progression enables us to trace how the cumulative layering of the three organizational capabilities shape the overall scaling capacity of a co-created sustainability solutions. For each case, we present: (1) an overview of the co-creation process, including key participants and the collaboratively defined problem and solution; and (2) an integrated assessment of how the presence or absence of each capability either facilitated or constrained subsequent scaling. This structure allows for both within-case and, subsequently in the discussion section, cross-case insights into how these conditions interact to shape scaling trajectories.

5.1. Rundgång, Sweden

Case description: The Rundgång project emerged within Karlstad Municipality as part of Sweden's national initiative "Climate Neutral Cities 2030″, launched in October 2021 under the Viable Cities program. Karlstad joined the network the same year through its local project, "Climate Neutral Karlstad 2030," with the goal of achieving municipal climate neutrality by 2030. At the same time, several local initiatives reflected a growing public interest in circular economy practices, including a 2021 motion by the Social Democratic Party to explore the establishment of a recycling mall, an e-petition from citizens, and a proposal from the Student Climate Council (elementary level) on textile recycling. In June 2022, the municipal council approved the motion and commissioned a feasibility study, in the form of a co-created experiment, which led to the creation of Rundgång, a recycling mall designed to promote circular economy principles. A municipal official explained that "the demand for something like this, a recycling mall, came from several directions, […] from politicians, from civil servants, from students in the student council, and from citizens through e-proposals (Municipal Official 1).

Throughout the remaining 2022 and the beginning of 2023, the project leaders situated in Karlstad Municipality's Environmental Office surveyed the local community to mobilize prospective participants. Subsequently, the mall opened to the public as a pilot project in October 2023, effectively tapping into and leveraging pre-existing local networks promoting sustainable consumption and cooperation among businesses and civil-society organizations. The coordination, however, was subsequently outsourced to an externally recruited project manager. Initially, core partners included the municipal energy company, the private property owner Mitt i City, Coompanion Värmland (representing the civil society), and over 40 small enterprises active in second-hand retail, repair services, upcycling, and smart-waste design. In particular, the impetus driving the widespread participation by small business owners was primarily value-driven, as it "nothing we can make a lot of money from, although we feel that this is where our heart is […], so that's why we are involved" (Small Business Owner 1).

Following the conclusion of the municipal pilot phase in late 2023, the initiative continued—albeit in a downscaled form—under the leadership of three small business owners who had participated during the pilot. Today, as of 2025, Rundgång comprises around twenty actors offering various recycling and circular services, including second-hand retail, remanufacturing, rentals, renovation, and repair. While the project successfully achieved its core objective of establishing a local hub for reuse and recycling, no formal plans were developed for upward institutionalization (scaling up) or replication (scaling out). Instead, the initiative has consolidated around

Table 2
The combination of organizational capabilities across the case studies.

Case	Stakeholder anchorage	Programmatic Integration	Resourcing capacity	Outward and upward scaling
SkyClean	Present	Present	Present	Present
Circular Green Blocks	Present	Partial	Partial	Absent
Arena Breivoll	Absent	Partial	Absent	Absent
Rundgång	Partial	Absent	Absent	Absent

Note: Absent = no indicators; Partial = one to two indicators; Present = three or more indicators observed (see Appendix 1 for operationalization of each network organizational capability).

a smaller and more stable constellation of stakeholders, shrinking from its initial forty participating private enterprises to about half that number.[4]

5.1.1. Integrated analysis

Rundgång involved broad stakeholder participation already during its initial pilot phase and showcased continued internal stakeholder anchorage based on its continued operations following the conclusion thereof and the subsequent transition towards smaller, locally consolidated operating model. This broad inclusion was an explicit strategic step taken, as it was recognized that "it was important to have a breadth of actors and representation of different types of actors" (Municipal official 1). The project successfully engaged a wide range of local actors during its design and implementation, all of whom shared a common goal of promoting circular economy practices and waste reduction, as clarified by representative for the property owner: "I see myself as a facilitator with a deep engagement and interest in sustainability" (Head of Property). However, local business engagement was largely bounded to the project's timeframe and geographically confined to Karlstad, suggesting a rather parochial outlook and commitment to the collaborative network. No explicit mechanisms were established to channel the stakeholder commitment beyond the pilot's localized context, nor were relational bridges built to external stakeholders—such as regional policymakers, national funders, or inter-municipal networks—that could have supported scaling upwards or outwards beyond the local context. As a result, the project exhibited internal, albeit not external, stakeholder anchorage that was insufficient for any further scaling ambitions.

Rundgång aligned with the broader municipal policy objective of achieving municipal climate neutrality by 2030, as also highlighted by the Climate Neutral Karlstad 2030 and Sweden's Viable Cities policy agenda, reflecting a generally supportive governance environment. The municipal council's decision to fund the feasibility study and launch the co-created initiative demonstrated programmatic alignment with local sustainability objectives, as Rundgång was "a direct measure connected to the municipality's climate goals, which wouldn't have been supported if that hadn't been the background context" (Municipal official 1). Yet, this programmatic integration remained shallow and temporary because the initiative was not subsequently embedded in longer-term municipal strategies, operational routines, or budgetary commitments that could have enabled continuation and broader expansion.

This limited institutional uptake was compounded by the absence of strategic leveraging by th Karlstad Municipality's Environmental Office in their capacity as network leaders. Although Rundgång was well positioned within wider policy agendas, the network leaders did not successfully capitalize on this alignment to build a stronger case for continued municipal commitment or to translate policy consistency into a concrete scaling pathway. The reason for this was the inability to think beyond the logic of a pilot experiment, which rendered the programmatic integration "less formalized; if it had been a long-term project, more explicit attention would have been directed at formalization and planning" (Municipal official 1). In consequence, Rundgång remained programmatically aligned with municipal policy agendas but only weakly integrated, meaning the initiative lacked the deeper institutionalization—mirroring its limited external stakeholder anchorage—necessary for durable programmatic integration and, ultimately, for scaling beyond its local context.

The project initially benefitted from a blended financing model that combined national (Viable Cities), municipal (Climate Neutral Karlstad 2030), and private contributions, including rent subsidies from the property owner. This mix effectively mobilized short-term resources to launch and operate the recycling mall. However, when the pilot phase concluded, public funding and formal coordination mechanisms were discontinued, leaving no stable framework for continued resource generation or network-wide financial governance. In practice, the municipality's limited long-term commitment—consistent with the initiative's shallow programmatic integration—meant that the project had to depend on revenue generated locally through patrons and participating enterprises, which constrained its capacity to invest in coordination, development, or expansion. This case illustrates how pilot-oriented design and modest scaling ambitions can inhibit the development of resourcing capacity over time, as neither stakeholders nor funders are incentivized to build longer-term financing arrangements or partnerships beyond the initial experiment.

Rundgång exemplifies a successful co-creation initiative without scaling capacity. The project's strong local engagement, clear purpose, and successful implementation demonstrate the potential of collaborative innovation at the municipal level. However, the partial presence of stakeholder anchorage, shallow programmatic integration, and weak resourcing capacity collectively constrained its potential to scale beyond Karlstad. The initiative thus remained primarily a localized experiment, valuable for generating learning and awareness, but without the institutional foundation or relational reach required to drive a broader sustainability transition. Nonetheless, there is a potential for scaling as there are indications of a broader interest outside the city as one informant observed how customers regularly would ask "could we have something like this where I live?" (Small business owner 1). Finally, the case also illustrates how co-creation initiatives may deliberately consolidate rather than expand. Faced with limited long-term municipal backing and no viable external resourcing pathway, the participating businesses effectively stabilized the initiative around a smaller, locally sustainable model. In effect, they dialed back scaling ambitions and instead prioritized maintaining day-to-day operations through self-financing and continued local patronage.

5.2. Arena Breivoll, Norway

5.2.1. Case description

Arena Breivoll was established in 2021 as an open, multi-actor collaborative platform aiming to co-create a shared vision and

[4] As outlined earlier, this may also variably be described as a form of scaling deep (Moore et al., 2015; Nair, 2021; Vreugdenhil et al., 2010), which for the purposes of this article does not qualify as the mode of scaling of analytical interest. See also endnote 2.

development strategy for transforming the Breivoll industrial area of Oslo into a sustainable, productive, and residential "micro-city" in alignment with the municipality's sustainability goals. The initiative was spearheaded by Pådriv Oslo, a not-for-profit social enterprise dedicated to promoting urban sustainability and collaborative innovation. Operating as a voluntary and experimental forum, Arena Breivoll sought to accelerate the city's planning processes and identify small-scale pilot investments to draw attention to the area's potential, including measures for biodiversity and green infrastructure preservation, with the long-term prospects of scaling upwards to the district- and city-level. A key motivating factor was that "the strategic city plan designated Breivoll as an urban area for radical functional mixing" (Project leader 1), which created an opportune moment to advance the co-creation project in the area. The platform was deliberately inclusive, inviting a wide range of stakeholders to participate in open workshops and dialogue meetings, such as local property owners and developers, businesses and tenants, creative industries, civil society organizations, academia, municipal departments (planning, environment, finance), regional and national transport agencies, and local politicians. Although participation varied, these interactions fostered early cross-sectoral engagement and public legitimacy. Meanwhile, Pådriv provided facilitative leadership as the convening and coordinating actor, supported by a small steering committee of committed stakeholders, including several private property owners and developers.

A catalytic grant from Oslo Municipality's Finance Department in 2021 funded the initial planning phase, and individual planning officials participated from the outset in deliberations over where pilots could be sited and how they could be designed experimentally. Their involvement strengthened institutional credibility by anchoring the collaboration in concrete, actionable planning work rather than aspirational visions. Moreover, the Planning and Building Agency held the formal prerogative—through zoning authority, permitting responsibilities, and related planning regulations—to translate proposals into implementable projects, making its engagement particularly consequential for moving from co-created ideas to realizable pilots. This process of co-designing pilot plans was seen among core participants as productive, as they managed to secure tentative support from prospective stakeholders to be involved in the pilots.

In 2022, the outputs of the project culminated with a set of shared plans for Breivoll's transformation, and the formulation of two short-term pilot projects proposed by dedicated working groups. However, neither of the two pilots were subsequently implemented or scaled. As financial resources from the private sector dwindled and municipal attention shifted, Arena Breivoll struggled to sustain momentum and interest in spearheading the pilot processes slowly dwindled. In effect, following months of inertia throughout the latter half of 2022, the project was formally discontinued in 2023. Despite its termination, the initiative demonstrated the potential of co-creative urban governance but also revealed the fragility of collaborative arenas lacking stakeholder anchorage and resourcing capacity.

5.2.2. Integrated analysis

Stakeholder anchorage in Arena Breivoll was present but fragile in the pilot phase. Pådriv succeeded in assembling a modest coalition of actors, but engagement was uneven and dependent on a few key individuals and stakeholders. This unevenness of commitment proved to be a source of fragility, as it meant that the convening, coordination, and problem-driving role was effectively agglomerated around Pådriv and the Planning and Building Agency, rather than shared among a broader set of participating stakeholders through collective initiative. Participation was thus more reactive than generative. Additionally, commitments were often personalized rather than institutionalized, as engagement relied heavily on a small number of individual relationships. For instance, one project leader described securing a property owner's involvement through a personal connection and shared interests "in ski jumping", noting that "touching their hearts will get people more engaged" (Project leader 2). In effect, Pådriv managed to get many stakeholders to sign off on the plans when invited, but none of them were proactively involved in the subsequent process of initiating activities, mobilizing resources, or assuming ownership of pilot implementation. Meanwhile, a parallel source of vulnerability was the minimal participation of the civil society, reflecting both the area's limited residential base and the experimental nature of the initiative. When the lead representative from the Planning and Building Agency fell ill and withdrew, momentum from the public administration quickly dissipated.

The subsequent withdrawal of a major property owner and developer triggered further disengagement among private actors. In parallel, the Transport authorities felt that the planning was premature, showcasing a lack of synchronization with local policy programs, which resulted in its withdrawal of active support. Despite the facilitative leadership of Pådriv and the inclusive principles guiding Arena Breivoll, the relational networks forged proved too shallow to compensate for the lack of durable institutional commitments by internal and external stakeholders. When attention shifted and key participants disengaged, the collaboration lacked the redundancy and shared responsibility needed to maintain momentum and to move from visioning to implementation. As a result, stakeholder anchorage subsequently eroded as multiple actors ended up "not being interested in the ongoing co-creation process" (Project leader 1).

Arena Breivoll initially demonstrated a partial form of programmatic integration, as its ambitions were explicitly framed within Oslo's broader climate and sustainability agenda and envisioned as a platform for linking pilot projects to subsequent implementation and scaling. In particular, Arena Breivoll could be an opportunity to explore and advance "Oslo's land use and transport plans, aligned with its focus on transit-oriented development and densification" (Planning official 1). In effect, the initiative aligned with the city's 2012 strategic plan for public space development, securing early support from the municipal planning and finance departments as well as modest start-up funding. However, this alignment gradually weakened due to the absence of complementary stakeholder anchorage and resourcing capacity. The launch of a parallel regional public transport planning process diverted municipal attention and resources, and key agencies began to perceive Arena Breivoll's proposals as premature given long-term investment timelines. Consequently, the initiative evolved from being a strategically embedded component of the city's sustainability planning to a largely informal and voluntary collaboration lacking both formal policy anchorage and a clear implementation pathway. Arena Breivoll

illustrates how, in the absence of mechanisms to negotiate across institutional silos and rigid planning hierarchies, co-creation initiatives risk being marginalized within existing administrative routines. Successful programmatic integration therefore requires both alignment with formal plans and, moreover, adaptive coordination capable of synchronizing scaling trajectories with higher-level institutional processes.

Resourcing capacity proved to be the decisive bottleneck. Beyond the initial catalytic grant from the municipal finance department, no long-term or scalable financial arrangements were secured, partly due to weak stakeholder anchorage. Although public agencies supported the project's vision, they anticipated that major investments would not begin before 2030, and the lack of detailed implementation plans as well as the withdrawal of support from several public stakeholders deterred private partners. In effect, the allocated funding was perceived by funders as a short-term contribution to an exploratory planning phase rather than as an investment anchored in a longer implementation trajectory. The project leader from Pådriv recalled an early meeting with a landowner who showed little concern for long-term commitments and instead responded glibly: "yes, this is interesting […] and that made basically them join and put money into [the project]" (Project leader 1). This interaction suggests that early private contributions were driven more by opportunistic interest and low-cost participation than by a shared understanding of the long-term resourcing requirements implied by implementation and scaling. As a result, the relationship to private funders remained shallow, as initial contributions did not translate into sustained financial commitments, nor did they create a credible funding pathway that could signal predictability to other partners. Pådriv's leadership later acknowledged that municipal actors must act not only as "planners but also as co-investors to create the predictability needed to attract [additional] funding" (Project leader 2). In the absence of such long-term public commitment, Arena Breivoll lacked the financial and institutional resources to move from planning to implementation.

The Arena Breivoll case demonstrates that even with some degree of programmatic integration, scaling capacity cannot develop in the absence of mutually reinforcing organizational capabilities. While Pådriv embedded scaling considerations in the form of a well-developed and clear vision for implementation and scaling, weak stakeholder anchorage and the subsequent collapse of resourcing capacity prevented this ambition from materializing. The absence of stable stakeholder commitment undermined efforts to mobilize sustained funding, effectively stalling the rollout of planned pilots. This dynamic illustrates how the presence of one capability, such as programmatic integration, cannot offset the absence of others. Once stakeholder anchorage eroded, resourcing capacity subsequently collapsed, and programmatic integration effectively lost its practical relevance.

5.3. Circular green blocks, Finland

5.3.1. Case description

The Circular Green Blocks (CGB) project was launched in 2021 under the leadership of the Helsinki Region Environmental Services Authority (HSY) in collaboration with Forum Virium Helsinki, Metropolia University, and Aalto University in an advisory capacity. Funded primarily through the EU's REACT-EU program and the City of Helsinki, the project had a total budget of €1.17 million. Its overarching aim was to promote circular and sharing economy practices in urban residential settings, particularly housing companies, by experimenting with concrete pilots such as shared spaces, yard development, and sustainable mobility solutions. Ten pilot experiments were designed between 2021–2022, the most visible of which was the e-cargo-bike sharing scheme in the Katajaharju neighborhood of Lauttasaari, Helsinki, launched in April 2023. The project sought to test how shared cargo-bikes, supported by a digital booking system and redesigned storage container, could offer residents a sustainable alternative to private car ownership. Between April and August, the residential end-users (from the Katajaharju neighborhood), CoReorient Oy (a private service provider of the cargo-bikes), HSY (in their capacity as network leaders), and representatives of the local housing organization were intensively involved in the co-created the design, contents, and features of the e-cargo-bike sharing scheme. The collaboration unfolded through an iterative process of collaborative problem-solving and prototyping, strategically scheduled during the Finnish summer, when local residents were expected to be most likely to use the e-cargo bikes as a potential substitution for their cars. The prototyping process involved continuous adaptation of both the software and the storage container, based on "a continuous input of digital and direct feedback and inputs from the stakeholders" (CoReorient Oy). Additionally, this feedback process was also supplemented by "direct and continuous conversations with local residents" (Property manager, Lautassari housing association) mediated by the housing association's property manager.

The pilot project was concluded in August 2023, after which its findings were codified in an informational booklet made publicly available to prospective service providers and end-users, such as local housing associations. Three tangible outputs included (1) four public guidelines for scaling (targeting housing companies, service providers, circular neighborhoods, and e-cargo-bike use); (2) 3D mapping tools to visualize how sharing solutions could be integrated at the block level; and (3) knowledge transfer to the City of Helsinki's circular economy roadmap and cluster program. While the project successfully delivered pilots and knowledge products, scaling beyond the pilot stage was limited. The e-cargo-bike service ended when the project terminated in September 2023, leaving its continuation indeterminate despite expressed interest in continuation by several citizens and the private service provider.

5.3.2. Integrated analysis

CGB developed robust internal stakeholder anchorage among its core partners (HSY, Forum Virium, Aalto University, and Metropolia University) as well as with local housing companies and the private provider CoReorient, which supplied and operated the e-cargo-bikes. A credible commitment to the co-creation process was signaled, in part, through the "contractual obligations signed by the stakeholders requiring a number of deliverables and continuous participation" (Project facilitator 1, HSY). Because the various actors were co-recipients of REACT-EU program, they were legally mandated to provide a range of deliverables as part of the project. However, the general level of engagement beyond the formal requirements was also reflected in the fact that the residents were not

passive beneficiaries, but proactively and eagerly engaged through iterative feedback loops, such as the redesign of the storage container following complaints about its aesthetics. External stakeholder anchorage extended further as multiple housing associations across Helsinki initially expressed interest in joining the pilot, demonstrating that the project resonated beyond the immediate neighborhood in Lauttassari. Moreover, these prospective stakeholders connected the initiative to Helsinki's broader urban policy ambitions on sustainable mobility and circular economy services. In consequence, the project exhibited a robust stakeholder anchorage, with credible external interest in scaling should enabling conditions materialize.

CGB also exhibited a partial presence of programmatic integration. The project linked indirectly to Helsinki's circular economy roadmap, ensuring some strategic alignment with municipal priorities. HSY, the network leader, expressed explicit scaling ambitions, framing the pilots as vehicles for codified knowledge that could be disseminated to other housing associations as "with ready-made templates for adoption" (Project facilitator 2, HSY). However, the depth of integration remained limited. Programmatic integration stopped at the level of knowledge production and dissemination, without concrete mechanisms to institutionalize pilots into the operational routines of housing associations or municipal planning. Informants highlighted a persistent gap between HSY's framing of pilots as temporary knowledge experiments and housing companies' preference for stable pathways toward implementation.

The reason for the weak scaling ambitions lay in its limited resourcing capacity, which was only partially present. CGB relied almost exclusively on short-term EU REACT-EU funding (90%) supplemented by City of Helsinki contributions (10%). No private investment was mobilized, and "no sustainable financing model was developed for continuation or expansion" (Project facilitator 2, HSY). While CoReorient benefitted from the project by showcasing its business model and gaining visibility, its involvement remained contractually bounded and temporary. With no sustained financial, human, or infrastructural resources beyond the pilot phase, resourcing capacity emerged as the critical bottleneck that prevented the project from scaling. This also had a negative feedback effect on the scaling ambitions in the programmatic integration, as HSY acknowledged from the outset that the available pool of funds would not suffice for any larger scaling efforts. The project thereby moderated its own scaling capacity due to its perceived limitations in resourcing capacity, despite its expressed interest to scale outwards and upwards.

CGB illustrates how despite the strong presence of stakeholder anchorage, programmatic integration was undermined by the limitations to resourcing capacity, both of which were only partially present. HSY had larger scaling ambitions in mind, but the lack of resourcing capacity did not provide further avenues to expand the initiatives outwards to other housing associations. In effect, they had to settle with the more moderate and feasible goal of producing publicly available informational pamphlets codifying the knowledge and experiences for other housing associations to freely replicate. Without the ability to command and sustain resources beyond the pilot, stakeholder anchorage could not be translated into broader adoption, nor could programmatic ambitions move beyond codified lessons. This configuration of network organizational capabilities explains why "the project produced valuable knowledge" (Project facilitator 3, HSY) but failed to achieve lasting expansion or institutional uptake.

5.4. SkyClean, Denmark

5.4.1. Case description

SkyClean was a co-creation initiative launched to enable substantial reductions in agricultural greenhouse gas emissions through the development and commercialization of a new breakthrough pyrolysis technology. The project's primary goal was to build a fully functional commercial pyrolysis plant and establish a sustainable business model to facilitate replication at scale, which was critically dependent on collaborating with prospective end-users of the pyrolysis plants, such as farmer cooperatives, agri-energy parks, or utilities companies, wanting to valorize straw and manure fibers for carbon-negative products. The initiative began in 2021 with the replication of a benchmark-model created by researchers at the Danish Technological University, leading to the subsequent construction of a small-scale testing facility that provided proof of concept. Following its technical success, the team constructed a medium-scale plant in 2022, ten times larger than the pilot, before completing a full-scale commercial facility in 2024 based on public funding from the government pyrolysis program. Stiesdal Fuel Technologies, led by the entrepreneur Henrik Stiesdal, acted as the project's network leader and principal investor. The company financed much of the early development and coordinated collaboration among a growing consortium of partners, including researchers from the Danish Technological University, engineers and technicians from Stiesdal's firm, the manager of the biochar factory in Vrå, local municipal officials, and Energy Cluster Denmark, which played an instrumental, intermediary role in securing public funding and regulatory support.

By 2024, SkyClean had achieved its immediate scaling objective with the establishment of a full-scale commercial pyrolysis plant, demonstrating the commercial and technical viability of integrating technological innovation, public-private collaboration, and business model design into a coherent scaling strategy. Plans to expand production to 50 additional plants across Denmark are underway, pending the resolution of regulatory barriers. With clear proof of concept and continuing institutional and financial support, full-scale commercialization is projected by 2027, positioning SkyClean as a benchmark for successful scaling of co-created sustainability solutions.

5.4.2. Integrated analysis

Internal stakeholder anchorage was deliberately narrow but strategically sufficient for providing proof of concept and achieving early scaling objectives. Rather than prioritizing broad participation, the initiative intentionally concentrated on assembling a small and highly competent coalition of actors with the technical expertise, financial capacity, and institutional credibility required to validate the pyrolysis technology and develop a viable business model. In parallel, farmers were included in testing the safety and logistics of spreading the biochar produced by pyrolysis on farmland. The core collaborative group worked closely and intensively together, characterized by mutual trust, complementary competences, and a shared commitment to demonstrating the technological

and economic feasibility of large-scale biochar production. This emphasis on aligning interests and demonstrating the viability of the business model is reflected in how one manager ceremoniously described stakeholder inclusion: "It may sound a little holy, but we believe that we cannot move forward before all the different stakeholders have an understanding of what we are doing" (Chief communications officer, Stiesdal).

The collaborative network was also reinforced by an explicit effort to build external anchorage with political and regulatory authorities. In effect, the public endorsement of three ministers was secured for the full-scale commercial facility inauguration in 2024, providing legitimacy and signaling strong governmental interest in the technology's potential to advance Denmark's climate neutrality goals. These credible endorsements helped smooth coordination across administrative levels and strengthened the project's credibility in the eyes of potential funders and investors. However, while political anchorage was robust, regulatory agencies were slower to adapt. The necessary policy frameworks for carbon accounting and biochar use in agriculture lagged behind technological progress, creating temporary barriers to full commercialization. Nonetheless, Stiesdal has anticipated that regulatory alignment is a matter of timing rather than opposition, thus planning a full commercial scaling by 2027.

Programmatic integration was a defining strength of SkyClean. The initiative's core mission—to demonstrate the technical feasibility and commercial viability of pyrolysis as a climate mitigation technology—was tightly coupled to Denmark's broader sustainability and energy transition agendas. In the backdrop of this sustainability transition agenda, scaling was not an afterthought but an explicit strategic objective embedded in the project design from the outset, aligning with the municipal plans and the organizational strategy of SkyClean. On the municipal level, Hjørring municipality has explicitly adopted a municipal plan that strategically bets on the potential synergy between existing biogas plants and the proliferation of pyrolysis technology, stating that they "want to ensure an interconnection between pyrolysis plants and biogas plants" (Climate chief, Hjørring Municipality).

At the organizational level, Stiesdal's leadership provided a unifying strategic framework that bridged experimental innovation, policy engagement, and business model development. Each successive facility (the pilot, mid-scale, and full-scale plants) served as a tangible anchor point for institutional learning and progressive scaling. This stepwise approach reflected a deliberate "learning-by-doing strategy" (CEO, Stiesdal) that combined technical experimentation with "the goal of developing a business model that makes it possible for all the involved parties—from farmers to investors—to build and operate new pyrolysis plants" (Chief communications officer, Stiesdal), effectively laying the groundwork for broader deployment. Through close coordination with its broad internal and external stakeholder anchorage, Energy Cluster Denmark and relevant ministries, the project actively sought to align its activities with state priorities and EU climate programs. This ensured that technological progress was matched by growing policy attention and legitimacy. Importantly, programmatic integration in SkyClean went beyond alignment with existing policies; it also entailed the creation of a scalable commercial model. The development of a cost-competitive production process for biochar and synthetic fuels provided the foundation for a viable pathway toward replication (scaling out) and investor appeal.

The project's resourcing capacity was equally robust, as Stiesdal's early investment provided the financial base and credibility necessary to initiate the pilot stages. This internal resourcing was progressively complemented by external sources as the project demonstrated success and reduced perceived risk. Energy Cluster Denmark played a critical intermediary role in coordinating public support, successfully securing a large state grant to finance the full-scale commercial plant. As reported by the chief research officer: "without this grant, SkyClean would not have come to the market" (Chief research officer, Stiesdal). In addition, SkyClean established a joint venture with biogas producers, securing access to organic feedstock, which is a crucial material resource that linked the project to existing circular economy infrastructures. This networked approach to resource mobilization exemplified dynamic resourcing capacity, whereby the collaborative network institutionalized resource flows and relationships rather than relying solely on discrete funding events. The project's ability to attract investors and policy support was deeply tied to its commercial viability. By demonstrating profitability metrics, such as cost efficiency, capital expenditure ratios, and measurable outputs, SkyClean reduced uncertainty for future investors. Although the government postponed releasing additional funds for subsequent plants pending final regulatory alignment, it allocated a dedicated funding stream for the pyrolysis industry, confirming strong institutional commitment to future scaling.

SkyClean demonstrates how the co-presence and mutual reinforcement of stakeholder anchorage, programmatic integration, and resourcing capacity create a virtuous cycle for successful scaling of SkyClean's experimental pilot project into a full-scale commercial plant. The project's narrow but targeted stakeholder anchorage among technically and financially capable stakeholders provided decisiveness and cohesion, which in turn enabled deep programmatic integration through a shared scaling vision and structured learning pathway. This integration enhanced credibility and positioned the project within national and EU sustainability agendas, thereby facilitating the attraction of resources and political backing. In this case, each capability amplified the others: (a) anchorage among competent and committed partners facilitated integration of innovation and policy; (b) integration strengthened the project's legitimacy and attractiveness to funders; and (c) resourcing capacity sustained the iterative expansion of technical and organizational infrastructures to form a viable business model for scaling.

6. Discussion

A cross-case analysis demonstrates that the three organizational capabilities—stakeholder anchorage, programmatic integration, and resourcing capacity—should be construed as mutually reinforcing components of scaling capacity. Each capability addresses a distinct and non-substitutable barrier to scaling, meaning that the absence of one creates a terminal bottleneck, regardless of the strength of the others. While the presence of a single capability may enable facets of scaling, only their combined and coordinated development enables collaborative networks to move from localized experimentation toward upwards and outwards scaled sustainability transitions (see Table 3).

Their interdependence is both functional, as each strengthens the others, and temporal, as the development of one capability often enables or accelerates the maturation of the rest. Deficits in any one capability have therefore shown to undermine the cumulative stability of the scaling process. This relational and processual understanding aligns with recent perspectives in collaborative governance, which emphasize that scaling capacity arises from the layering and integration of interlocking governance capabilities rather than from isolated managerial interventions (Chen and Hustad, 2025). This processual–relational view furthermore cautions against treating aspects of collaborative governance as discrete and isolated variables, consistent with the findings across the four case studies, instead emphasizing how their cumulative layering stabilizes and sustains specific collaborative dynamics aimed at scaling.

These findings extend on the extant literature in several respects. The broader scaling and social innovation literature has tended to conceptualize the progression from pilot to scale as a phased, largely sequential process in which enabling conditions operate additively: secure resources, demonstrate proof of concept, attract adopters (Seelos and Mair, 2017; Westley and Mcconnell, 2010). Our cases suggest a more recursive dynamic. The three capabilities do not simply accumulate in sequence; rather, they condition each other's development, such that weakness in one can retroactively erode the others. This configurational logic resonates with the emphasis in strategic niche management on the co-evolution of expectations, networks, and learning processes within protected spaces (Kemp et al., 1998), but adds a further layer of qualification by identifying the organizational capabilities through which such co-evolution is enacted in collaborative governance settings. Three points of departure from the existing literature merit particular attention.

First, stakeholder anchorage forms the relational and legitimacy foundation upon which the other two capabilities build. Well-anchored networks command greater commitment, trust, and credibility among key stakeholders, which not only sustains participation but also increases the likelihood that stakeholders will follow through on their roles when scaling requires concrete action. In this sense, anchorage is not merely interpersonal cohesion; it also entails clarity about responsibilities, sequencing, and formal or informal levers of influence through which network leaders and core stakeholders can coordinate implementation and hold partners to their commitments as the initiative expands (Ansell et al., 2022). This finding further qualifies the collaborative governance literature's treatment of stakeholder engagement, which has predominantly focused on the conditions for initiating and sustaining participation within a given collaborative arena (Emerson and Nabatchi, 2015).

The cases demonstrate that a qualitatively different form of stakeholder anchorage is required for scaling, that is, one that extends beyond the internal cohesion of the original coalition to encompass relational bridges to external actors whose endorsement, resources, or jurisdictional authority are prerequisites for expansion. The cases suggest that external anchorage functions through credible demonstrations of shared risk and mutual benefit, as illustrated by SkyClean's deliberate construction of a commercially viable business model that gave prospective partners concrete reasons to commit. At the same time, stakeholder anchorage also strengthens resourcing capacity because predictable resource access in co-created settings often depends on relational confidence and legitimacy rather than purely contractual, market-based financing logic (Warsen et al., 2019). Funders and resource partners are more likely to commit and renew support when they perceive a credible collaborative setting in which roles are upheld, and costs are not likely to be redistributed if partners disengage.

Strong stakeholder anchorage also facilitates programmatic integration by aligning diverse actors around shared strategic objectives and by ensuring that influential stakeholders are willing to translate collaborative ambitions into plans, routines, and decision processes within their respective organizations (Menon et al., 2024). This is illustrated negatively by both Rundgång and Arena Breivoll. In Rundgång, stakeholder anchorage remained largely parochial: local businesses were willing to participate in and sustain a locally bounded initiative, but had limited incentives or ambition to commit to scaling beyond Karlstad. As a result, the coalition lacked the outward-facing alignment and commitment needed for programmatic integration into broader strategies or inter-municipal pathways, and there was little basis for mobilizing external resources for expansion. In Arena Breivoll, commitments were not anchored in a shared understanding of the longer-term scaling trajectory, and when the coordination demands of implementation became salient, support weakened.

Second, programmatic integration, in turn, reinforces both stakeholder anchorage and resourcing capacity by institutionalizing scaling intentions within durable structures. Embedding an innovation into strategic plans, sectoral policies, or performance systems signals long-term commitment and stability, deepening stakeholder trust and legitimacy while creating predictable frameworks for funding and resource allocation (Lam et al., 2020). To this end, an integral feature of successful programmatic integration is embedding an explicit vision of scaling in the collaborative network (Nair, 2021), as it ensures that stakeholders can be strategically aligned and mobilized towards the realization of such scaling ambitions. In effect, this institutional embedding transforms temporary enthusiasm into sustained engagement and makes scaling a recognized organizational priority rather than an ad hoc aspiration (Vedeld et al., 2021).

These dynamics are illustrated by the SkyClean case, where scaling was embedded from the outset in a structured program—moving deliberately from a proof-of-concept facility to a medium-scale plant and then to a full-scale commercial installation. Stakeholders could coordinate around a shared roadmap with defined milestones, which strengthened commitment and increased funder confidence. This stepwise trajectory mirrors the phased logic described by Seelos and Mair (2017), but with an important difference: in SkyClean, progression between phases was not driven by a single organization's strategic decisions but by the collaborative network's capacity to renegotiate commitments and adapt governance arrangements at each stage. A contrasting illustration is provided by the Circular Green Blocks. Although the project was strategically aligned with Helsinki's circular economy ambitions and produced codified outputs (guidelines and tools), this alignment largely remained at the level of knowledge transfer rather than being translated into binding operational routines, budget lines, or adoption mandates within housing companies or municipal units. In the terminology of the policy experimentation literature, CGB exhibited the common pattern whereby pilot success under exceptional conditions fails to generate the institutional embedding required for broader adoption (van Buuren and Loorbach,

Table 3
Terminal bottlenecks created by the absence of network organizational capabilities.

Condition missing	Why scaling fails	Exemplars
Stakeholder Anchorage	• Primary mechanism: lack of legitimacy and political support generates resistance and inhibits adoption in new contexts. Furthermore, narrow or project-bound engagement erodes continuity beyond the pilot phase, leading to stakeholder disengagement once initial objectives are met. • Corollary 1: Undermines programmatic integration as weak or absent stakeholder commitment blocks ambitions and plans for scaling, as no broad coalition exists to support scaling. • Corollary 2: Undermines resourcing capacity as it limits access to funding channels and reducing credibility among potential investors or partners.	Arena Breivoll shows how hte withdrawal of pivotal public-sector champion and major private actor triggered cascading disengagement; limited civil-society anchorage and weak external commitments reduced the initiative's ability to mobilize follow-on support.
Programmatic Integration	• Primary mechanism: no strategic or institutional vehicle exists to carry scaling forward once the initial project ends. In effect, failure to design for scaling (e.g., through proof-of-concept and institutional embedding) diminishes credibility and reduces the ability to mobilize future resources. • Corollary 1: Undermines stakeholder anchorage as the absence of an overarching vision or framework weakens long-term stakeholder commitment and undermines collective direction. • Corollary 2: Undermines resourcing capacity as scaling efforts remain fragmented and lack a clear vision, which discourages prospective stakeholders or funders to pledge their support and/or resources	Rundgång was treated as a pilot feasibility experiment with limited institutional embedding, hence there no municipal routinization or formal scaling pathway. Ultimately, continuation (scaling deep) depended on a reduced set of entrepreneurial actors rather than organizational integration.
Resourcing Capacity	• Primary mechanism: scaling stalls due to the lack of financial, material, or organizational means to expand beyond the pilot stage. Scaling ambitions becomes constrained as even well-designed and legitimate initiatives cannot progress without stable financial and human capital foundations. • Corollary 1: Undermines stakeholder anchorage as it reduces incentives for long-term participation and thus access to potential resource and knowledge flows • Corollary 2: Undermines programmatic integration as, without adequate resourcing, scaling plans and ambitions remain locally isolated initiatives.	Circular Green Blocks illustrates how heavy dependence on time-limited external funding, limited private co-investment lead to the termination of its pilots, despite demonstrable external interest and strong local engagement.

2009). What the case adds to this account is that the gap was not simply a failure of knowledge transfer, but a consequence of constrained resourcing capacity that preemptively narrowed the network's own scaling ambitions.

Finally, resourcing capacity enables anchorage and integration to translate into sustained and adaptive action. Without sufficiently predictable financial, material, or human resources, stakeholder commitment is difficult to maintain, and even well-integrated initiatives risk stalling at the point where plans must be converted into implementation. This is visible across the less successful cases. In Arena Breivoll, the absence of follow-on funding after the initial catalytic grant pushed partners into a "wait-and-see" mode; in Circular Green Blocks, the termination of REACT-EU financing ended the e-cargo-bike service despite expressed interest in continuation.

The cases also show that resourcing capacity shapes stakeholder anchorage through expectations. When the resource implications of scaling are, for one reason or another, underestimated or not communicated early, scaling demands can trigger disappointment and withdrawal as actors realize that additional costs, risks, or workloads will fall unevenly on those who remain engaged. In this sense, weak or uncertain resourcing can produce cascading, effects (Mclaughlin et al., 2022), as disengagement by some stakeholders increases the perceived burden on others, further eroding commitment and undermining collective ownership. This dynamic problematizes and extends the cognate concept of mobilization capacity (Menon et al., 2024), which theorizes how actors extract resources from their networks, by specifying a feedback mechanism that operates in the opposite direction. In essence, they argue that when resource pathways prove fragile, the relational infrastructure through which mobilization occurs can itself deteriorate. The implication is that resourcing capacity is not merely a downstream consequence of network strength but actively constitutes and reconstitutes the relational conditions on which anchorage depends. SkyClean illustrates the opposite dynamic, inasmuch as the deliberate construction of a credible business model and staged scaling pathway reduced uncertainty for partners and funders, helping sustain commitment and providing the slack needed to adapt arrangements as the initiative expanded (Rivera et al., 2025).

The four case studies furthermore allow us to revisit the timing question on whether scaling is required as a deliberate, upfront strategy that must be planned in advance, or if it can be negotiated ad hoc and incrementally even when the pilot phase of co-creation process has concluded. The case studies show that a reflexive and deliberate scaling ambition is closely connected to the successful cultivation of the three network organizational capabilities. In essence, if a collaborative network has no explicit ambition or strategy to scale, the institutional and relational capacities that enable scaling appear unlikely to materialize spontaneously.

Thus, while scaling ambitions might nonetheless be successfully introduced in later stages of collaborative governance, the delay is usually a liability rather than an asset to its realization. Without deliberate intent and leadership to manage this transition, collaborative networks exhibit a predilection for consolidating around their initial achievements rather than expanding their reach. This is

illustrated by Rundgång, where stakeholders consolidated the initiative locally once it became clear that viable scaling pathways were not in place. Moore et al. (2015) describe this trajectory as "scaling deep," whereby initiatives intensify their impact within existing boundaries rather than extending outward or upward. Our cases indicate that scaling deep is often not a proactive strategic choice but a default that collaborative networks gravitate toward when the organizational capabilities for outward or upward scaling cannot be assembled. More generally, then, scaling capacity is best understood as a strategic organizational investment that must be deliberately envisioned and developed rather than expected to emerge organically from collaboration alone. Where the underlying network organizational capabilities cannot be built and sustained, attempts to scale up or out may be strategically inexpedient, and local consolidation may represent a more realistic and valuable trajectory.

7. Conclusion

This article has developed and empirically illustrated a capability-based framework for understanding how co-created sustainability solutions can be successfully scaled beyond their initial project contexts. We have argued that scaling capacity in collaborative governance networks rests on the combined and interdependent presence of three network organizational capabilities—stakeholder anchorage, programmatic integration, and resourcing capacity—each addressing distinct and non-substitutable barriers to scaling. The cross-case comparison of four Nordic co-creation initiatives demonstrates that these capabilities are best understood not as isolated determinants, but as mutually reinforcing components of a cumulative scaling capacity that can be mutually reinforcing when co-present or mutually constraining when one or more capabilities are absent. Their effectiveness depends on their functional interplay and temporal sequencing, where the absence of one capability can undermine the others, producing terminal bottlenecks that limit the potential for sustainable institutionalization and broader transition impact. Theoretically, this study contributes to the literature on innovation scaling, collaborative governance, and sustainability transitions by advancing a cumulative perspective on governance networks (Chen and Hustad, 2025). Whereas previous research has often treated leadership, resources, and institutional embedding as separate enabling factors (Torfing et al., 2024), this framework conceptualizes them as interlocking organizational capabilities that evolve through iterative, co-dependent processes. This cumulative view helps clarify how scaling emerges from the gradual layering of governance capacities rather than from discrete managerial interventions.

The findings underscore that achieving scaling in co-created sustainability initiatives requires deliberate leadership and long-term strategic foresight (Hofstad and Vedeld, 2021; S. Page, 2010; Sørensen et al., 2021). Facilitative leaders must not only manage collaboration effectively but also cultivate the capabilities that enable scaling capacity, namely securing durable stakeholder anchorage, embedding co-created solutions within institutional frameworks, and developing sustainable resource flows. A holistic leadership perspective is thereby needed, insofar as the combined and balanced development of stakeholder anchorage, programmatic integration, and resourcing capacity is integral to build lasting scaling capacity.

Several limitations should nonetheless be acknowledged. Although the comparative design controls for contextual variation by focusing on the shared Nordic governance environment—characterized by high trust, decentralization, and collaborative traditions—the study examines only four of the 27 possible configurations of the three capabilities.[5] Other constellations of presence and absence might reveal different pathways, tensions, or compensatory mechanisms affecting scaling outcomes. Future research based on comparative and set-theoretical approaches, for example, fuzzy set qualitative comparative analysis, can expand the empirical scope to include diverse governance contexts and additional case combinations to further test the robustness and generalizability of the framework (Molenveld et al., 2021; Øjvind et al., 2024; Torfing et al., 2020).

Despite these limitations, the study provides evidence for how scaling can be systematically analyzed as a function of interdependent organizational capabilities rather than as a byproduct of isolated factors. By demonstrating this within the methodologically controlled Nordic setting and across cases of successful co-creation, we offer both theoretical clarity and practical guidance for advancing sustainability transitions. In conclusion, the article lays the groundwork for a more integrated research agenda on the dynamic capabilities of collaborative governance networks, aimed at understanding how co-created innovations can move from local experiments to transformative, scalable changes.

CRediT authorship contribution statement

Alexander L.Q. Chen-Florea: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Conceptualization. **Mikael Granberg:** Writing – review & editing, Validation, Investigation, Data curation. **Hege Hofstad:** Writing – review & editing, Visualization, Validation, Investigation, Data curation. **Trond Vedeld:** Writing – review & editing, Validation, Investigation, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

[5] Each of the three organizational capabilities—stakeholder anchorage, programmatic integration, and resourcing capacity—can take one of three values (absent, partially present, or present), yielding a total of 27 possible configurations ($33 = 27$).

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.eist.2026.101167.

Data availability

The research protocol based on which data has been collected and supplementary case reports can be accessed at www.gogreen-project.com.

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